ELECTROMAGNETISM: ANSWERS

This is a closed-book examination. A formula sheet is provided

Question 1. [a) = 5] [b) = 5] [c) = 10] [d) = 5] [e) = 10] [f) = 5]

a) (bookwork) In free space $\rho = 0$, $\mathbf{J} = 0$, $\mathbf{D} = \varepsilon_0 \mathbf{E}$, $\mathbf{B} = \mu_0 \mathbf{H}$ [2]. Harmonic time variation implies $\partial/\partial t = j\omega$ [1], so that Maxwell's equations are [2]

$$\nabla \times \mathbf{E} = -j\omega\mu_0 \mathbf{H}$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon_0 \mathbf{E}$$

$$\nabla \cdot \mathbf{E} = 0 \qquad \nabla \cdot \mathbf{H} = 0$$

b) (calculated example) For static electric fields $\nabla \times \mathbf{E} = 0$ [2]. For the expression given $\nabla \times \mathbf{E} = \sin x \sin y \mathbf{e}_z \neq 0$ [3], which does not satisfy Maxwell's equation.

c) (calculated example) The two Maxwell's equations for static magnetic fields are $\nabla \times \mathbf{H} = \mathbf{J}$ [2], and $\nabla \cdot \mathbf{H} = 0$ [3]. For the field given, we obtain $\nabla \cdot \mathbf{H} = 0$, which satisfies one of the equations [2]. The other equation gives

$$\mathbf{J} = \mathbf{e}_x + \mathbf{e}_y + \mathbf{e}_z \qquad [3]$$

d) (calculated example) For a quarter-wave transmission line of a length d, we have

$$kd = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2} \qquad [2]$$

so that $\tan(kd) = \infty$ and

$$Z_{\text{in}} = \frac{Z_0^2}{Z_{\text{L}}} \qquad [3]$$

e) (calculated example) The transmission lines are matched to the ports, and therefore there will be no reflection from the interfaces between the ports and the lines [1]. For the same reason, the S-parameters referred to the two-port device will be the same for cases (a) and (b) [1]. A wave travelling along a transmission line will acquire a phase-shift e^{-jkd} [1]. The physical meaning of the S-parameters are voltage reflection and transmission coefficients [1]. Therefore, the values of the new S-parameters will be

$$S_{11} e^{-j2kd_1} \qquad [3]$$

and

$$S_{21} e^{-jk(d_1+d_2)} \qquad [3]$$

f) The relationship is $\omega = ck$ [2], where c is the wave velocity [1]. It is the velocity with which the phase front of the wave propagates [2].

Question 2. [a) = 15], [b) = 5], [c) = 10]

a) (bookwork) An impedance Z is a one-port device, and a VNA connected to it will be able to measure S_{11} [2]. Using an expression from the formula sheet, and bearing in mind that $Z = V_1/I_1$ [3], we get

$$S_{11} = \frac{Z - Z_0}{Z + Z_0} \qquad [5]$$

Rearranging, we get

$$Z = Z_0 \frac{1 + S_{11}}{1 - S_{11}} \qquad [5]$$

so that the complex value of the impedance can be determined from a measurement of S_{11}.

b) (reasoning) there are many possible answers, for example:

advantage: high frequency range, ability to determine the impedance magnitude and phase [2] disadvantage: high cost, low accuracy for large and small impedances. If the VNA impedance and the measured impedance are significantly mismatched, then $S_{11} \approx 1$ or -1, and very large and very small impedances are difficult to measure accurately. [3]

c) (calculated example/reasoning)

The device is matched to port 1, therefore $S_{11} = 0$ [2]. For incidence from port 1, the voltage at port 2 is twice the input voltage. Therefore $S_{21} = 2$ [3].

For incidence from port 2, the voltage source is connected to the ground, and the device presents a short circuit. Therefore $S_{22} = -1$ [3]. There is no transmission from port 2 to port 1, therefore $S_{12} = 0$ [2].

Question 3. [a) = 5], [b) = 10], [c) = 5] [d) = 10]

(all questions are bookwork)

a) The assumption of z-going x-polarised wave means that $\mathbf{E} = E(z)\mathbf{e}_x$ [3]. Then the vector wave equation takes the form

$$\frac{d^2E(z)}{dz^2} + \omega^2\mu_0\varepsilon_0\left(1 - j\frac{\sigma}{\omega\varepsilon_0}\right)E(z) = 0 \qquad [2]$$

b) Assuming $E(z) = E_0e^{-jkz}$ [2], the scalar wave equation takes the form

$$-k^2 + \omega^2\mu_0\varepsilon_0\left(1 - j\frac{\sigma}{\omega\varepsilon_0}\right) = 0 \qquad [2]$$

leading to

$$k = \omega\sqrt{\mu_0\varepsilon_{\text{eff}}} \qquad [4]$$

where

$$\varepsilon_{\text{eff}} = \varepsilon_0\left(1 - j\frac{\sigma}{\omega\varepsilon_0}\right) \qquad [2]$$

c) The assumption of high conductivity means that

$$1 - j\frac{\sigma}{\omega\varepsilon_0} \approx -j\frac{\sigma}{\omega\varepsilon_0} \qquad [2]$$

leading to

$$k = (1-j)\sqrt{\frac{\omega\mu_0\sigma}{2}} \qquad [3]$$

d) The wavenumber is complex $k = k' - jk''$ [2], and the wave amplitude decays as $\exp(-k''z)$ [3]. Skin depth is the distance δ at which the amplitude decays by e times, namely $\delta = 1/k''$ [3], giving

$$\delta = \sqrt{\frac{2}{\omega\mu_0\sigma}} \qquad [2]$$